

Physiology of Cardiac Cycle – Part 2 – Loops, Laws and Cardiac Output

PPOM 2 - Cardiovascular System

Vishwa Rajagopalan, PhD, FAHA, FCVS

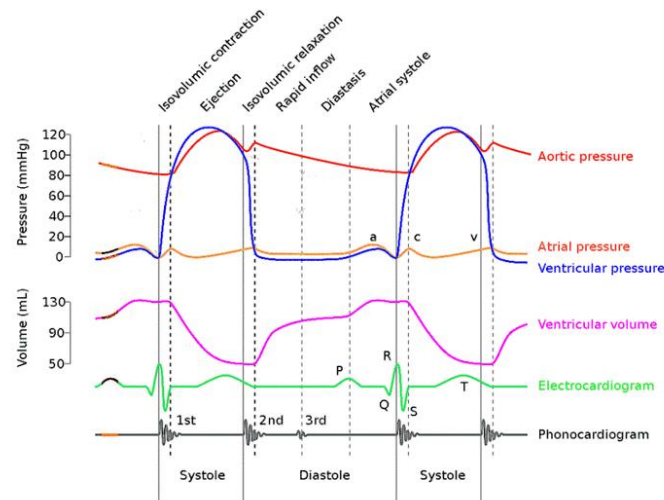
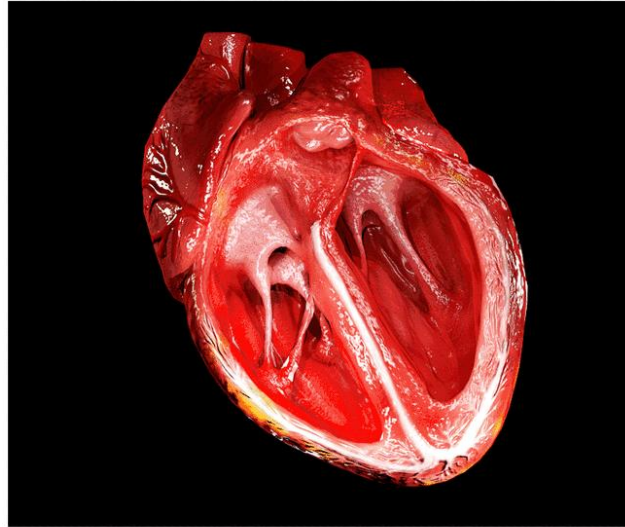
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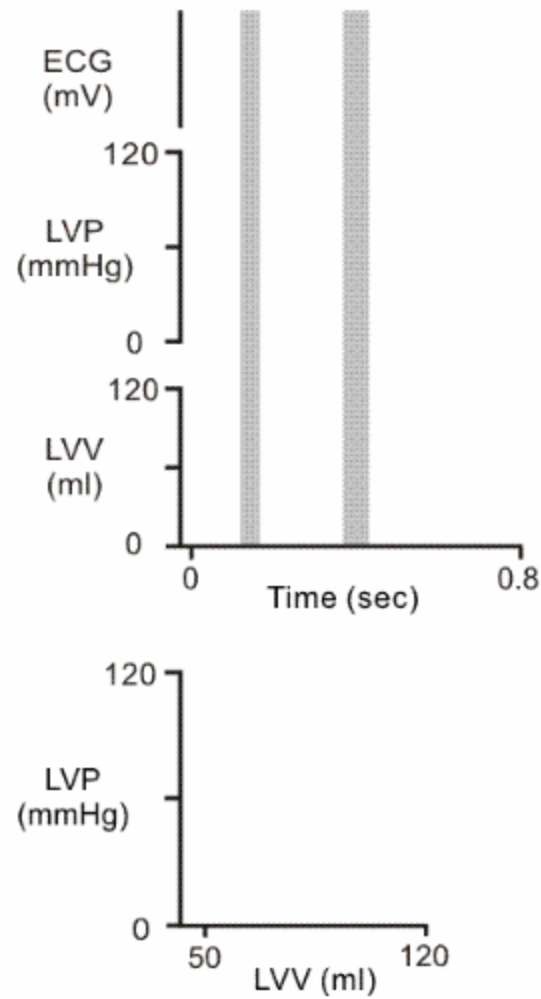


Session Objectives

1. Graph the relationship between cardiac pressures and volumes using the Law of Laplace
2. Recognize concepts of cardiac output and stroke volume in a cardiac cycle
3. Discriminate how afterload, preload (Frank-Starling's Law), and contractility regulate stroke volume and cardiac output in cardiac cycle
4. Apply Fick principle in the measurement of cardiac output
5. Extrapolate the phases, loops, laws, and cardiac output regulation in representative cardiovascular pathologies

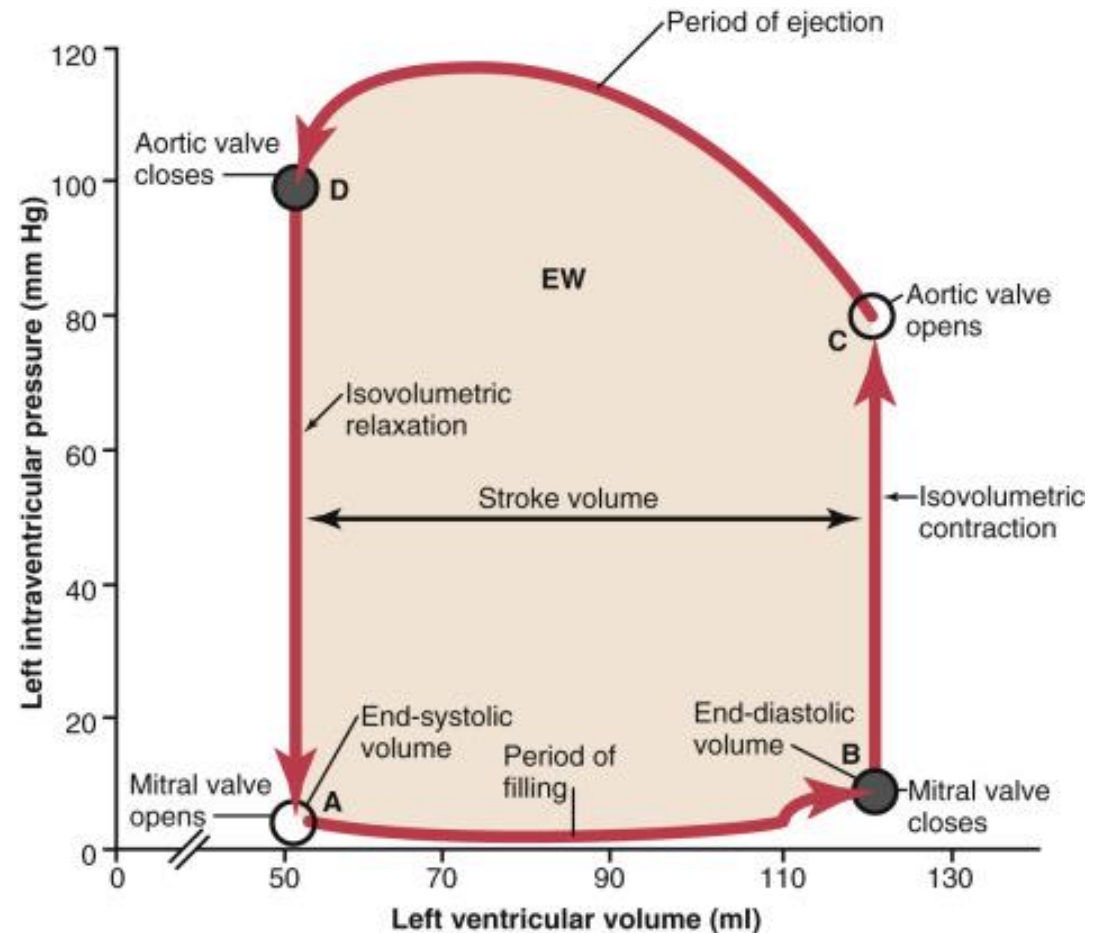
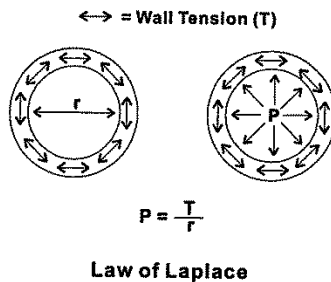
CARDIAC CYCLE





Pressure-Volume Curve and Laplace Law

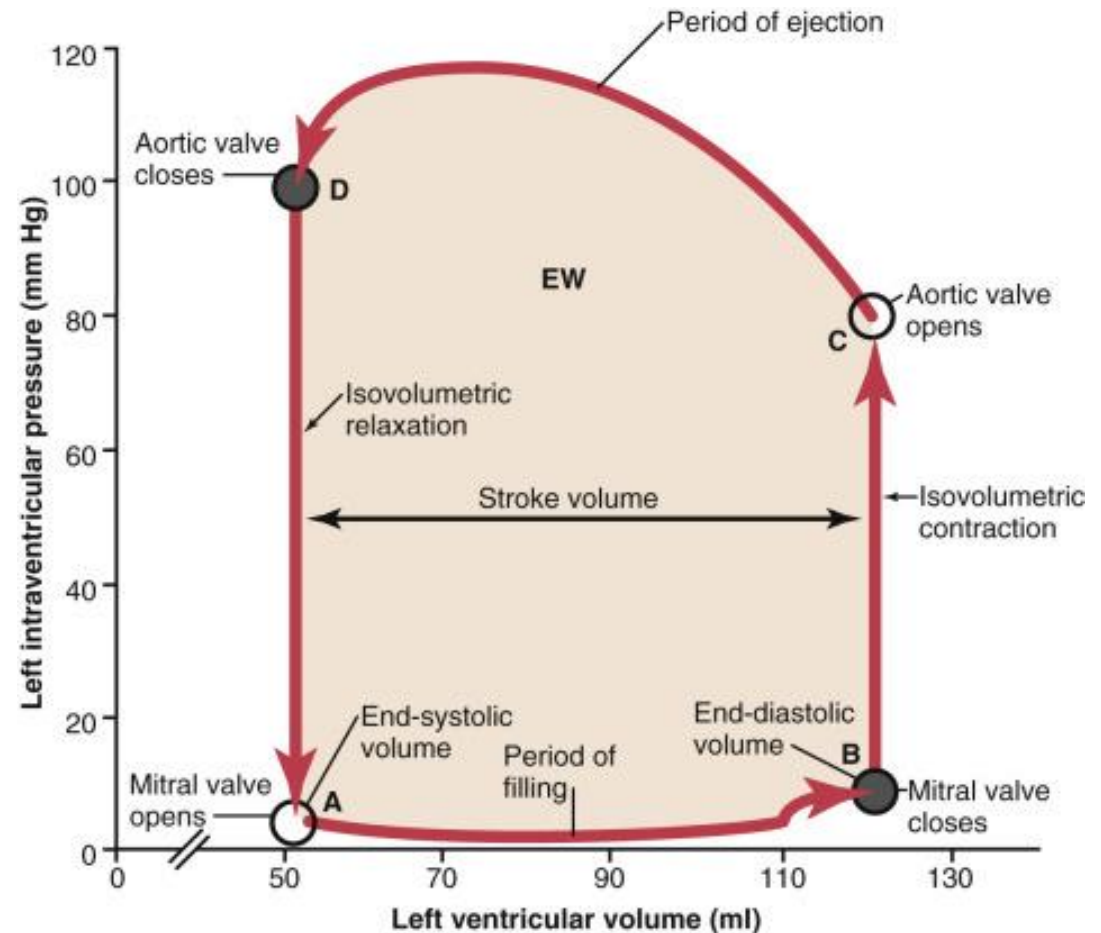
- LAW OF LAPLACE** relates **ventricular pressure (P)** to the **tension (T)** in the ventricular wall divided by the **ventricular radius (r)**.



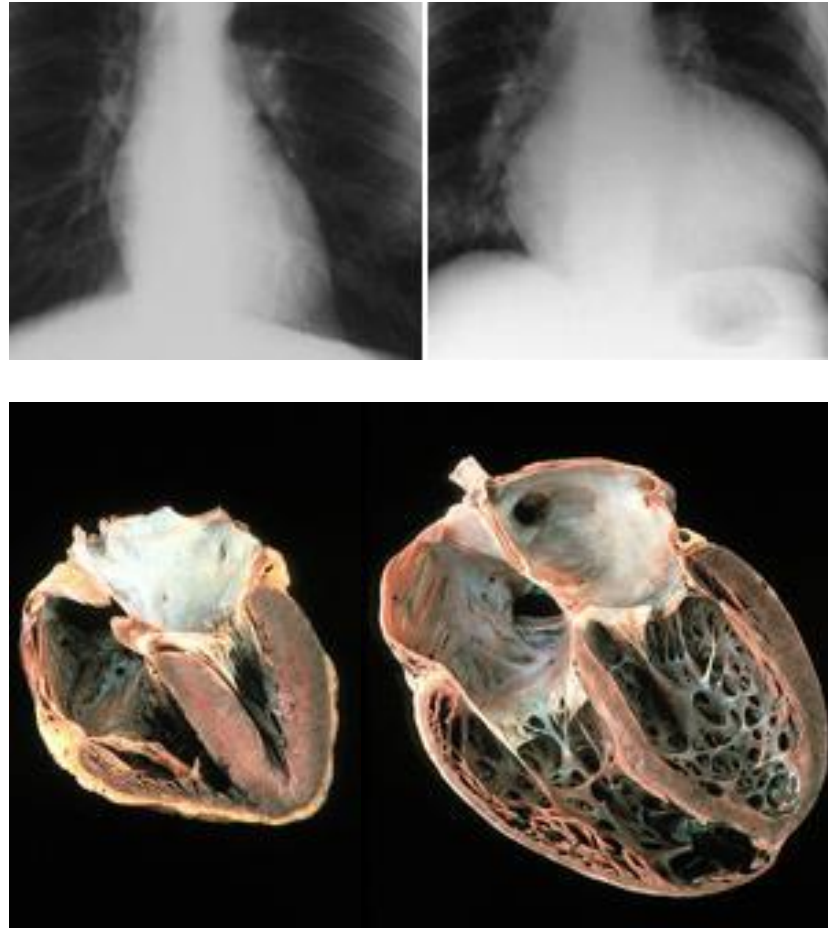
Pressure-Volume Curve and Laplace Law (contd...)

- The **end diastolic volume (EDV)** is referred to as **preload**.
- The **afterload** is proportional to **arterial pressure**

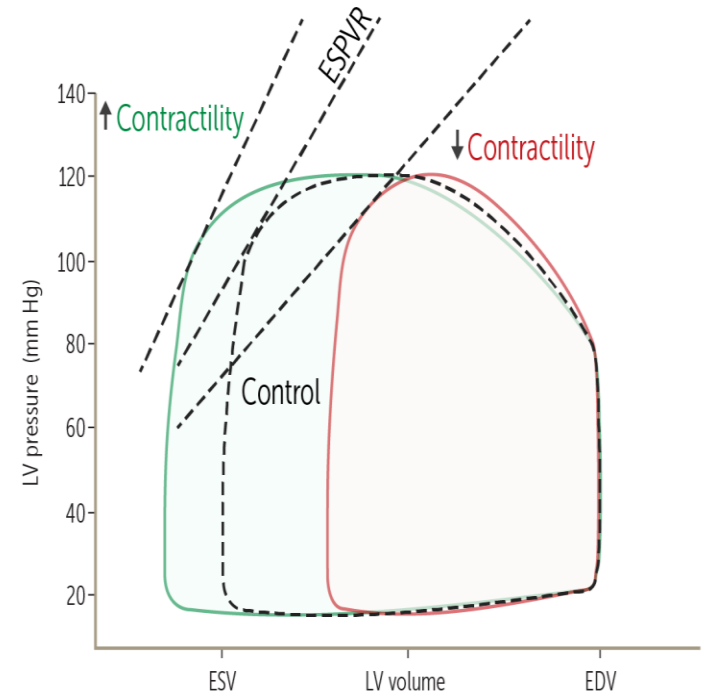
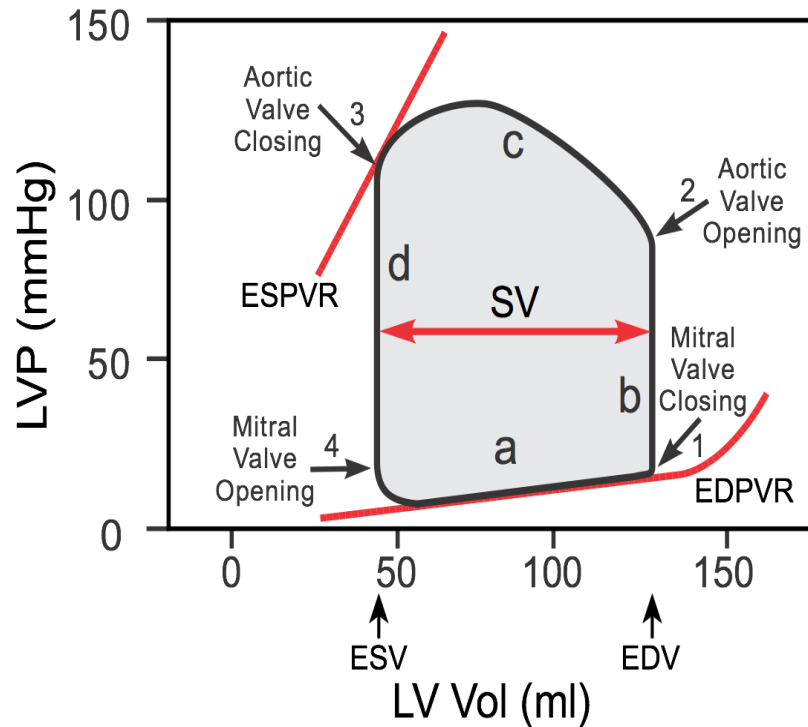
$$\sigma \propto \frac{P \times r}{h}$$



Dilated Cardiomyopathy



Ventricular Pressure-Volume Relationship



Stroke Volume and Cardiac Output

- The amount of blood ejected by the heart in each beat is the **stroke volume** (SV).

$$\mathbf{SV \text{ (ml)} = EDV \text{ (ml)} - ESV \text{ (ml)}}$$

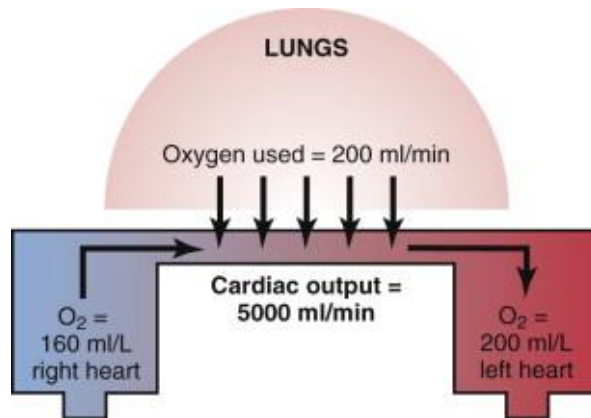
- Cardiac output** is the amount of blood pumped by the heart in a given amount of time.

$$\mathbf{CO \text{ (ml/min)} = SV \text{ (ml/beat)} \times HR \text{ (beats/min)}}$$

- At an SV of 80ml/min, the CO is 5,760 ml/min ($72 \times 80 = 5,760$).

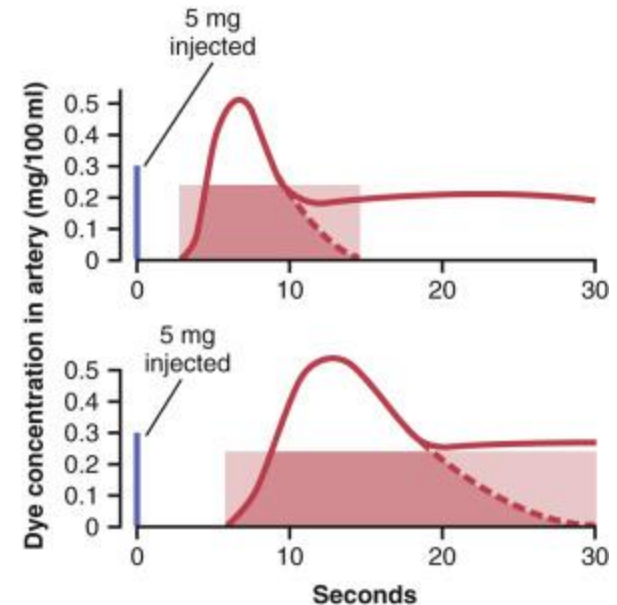
Cardiac Output measurement

Fick principle



$$\text{Cardiac output (L/min)} = \frac{\text{O}_2 \text{ absorbed per minute by the lungs (ml/min)}}{\text{Arteriovenous O}_2 \text{ difference (ml/L of blood)}}$$

Indicator dilution

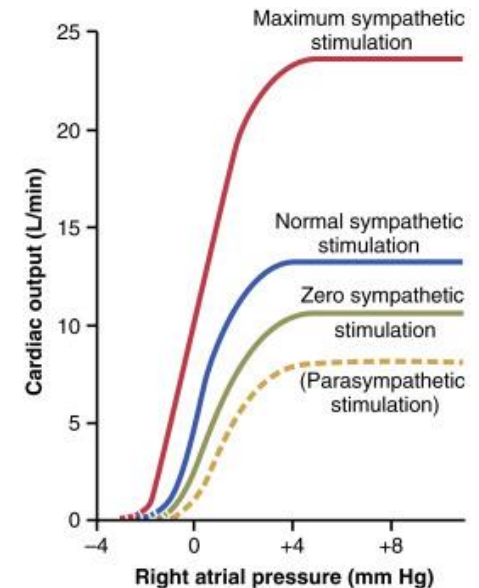


$$\text{Cardiac output (ml/min)} = \frac{\text{Milligrams of dye injected} \times 60}{\left(\text{Average concentration of dye in each milliliter of blood for the duration of the curve} \right) \times \left(\text{Duration of the curve in seconds} \right)}$$

Regulation of HR and SV and CO

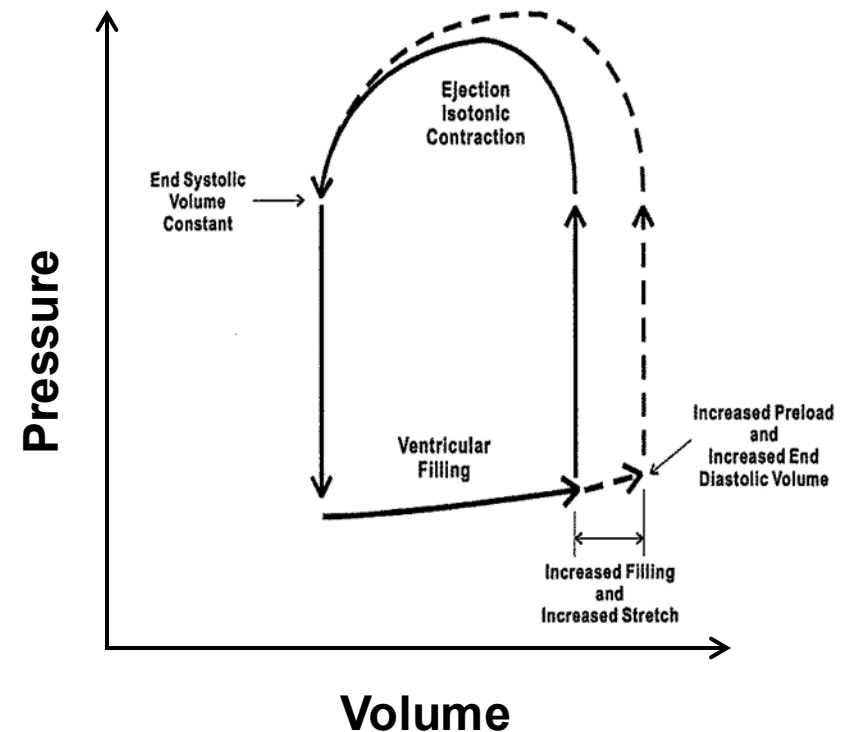
HR can be changed by sympathetic and parasympathetic stimulation.

- Increase in HR decreases SV (decreases the amount of time the heart spends in diastole) and thus the CO is maintained.
 - However, as HR increases to about 180-200 bpm, CO increases. This is because the increase in HR more than compensates for the decrease in SV.
 - Beyond this range both stroke volume and CO decrease.
- Goal in changing HR and/or SV is to aid in **maintaining the mean arterial pressure (MAP)**. **Constriction of arterioles and veins** also aid in maintaining BP.
 - **MAP = CO x TPR (Total peripheral resistance)**



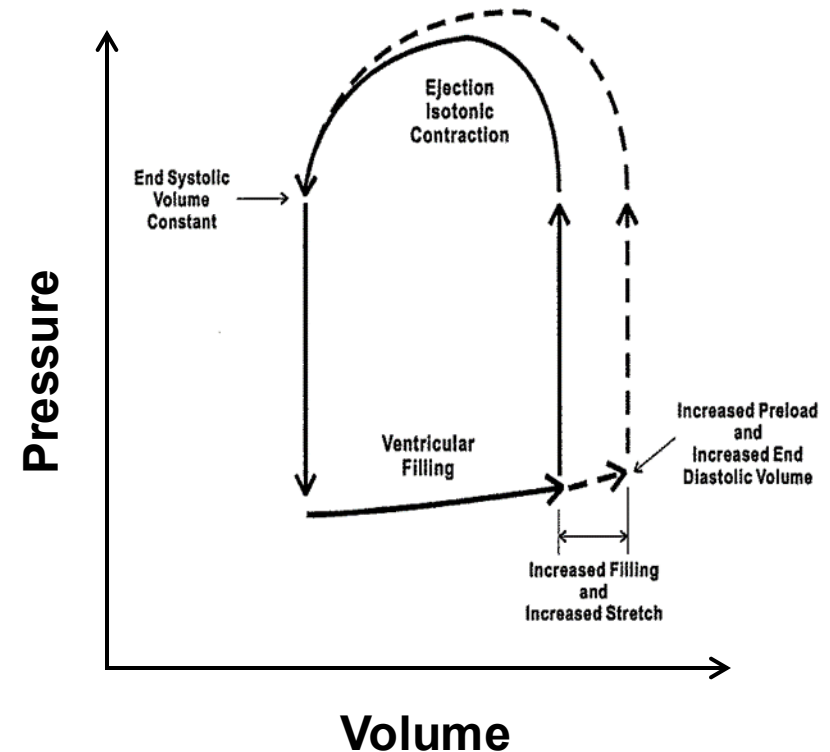
Frank-Starling Law – Effect of Preload (contd)

- The **FRANK-STARLING LAW** (Starling's Law of the Heart) states that SV increases as ventricular EDV increases.
- The greater the passive stretch or preload of the ventricle, the greater the force of ventricular contraction, and the greater the SV.

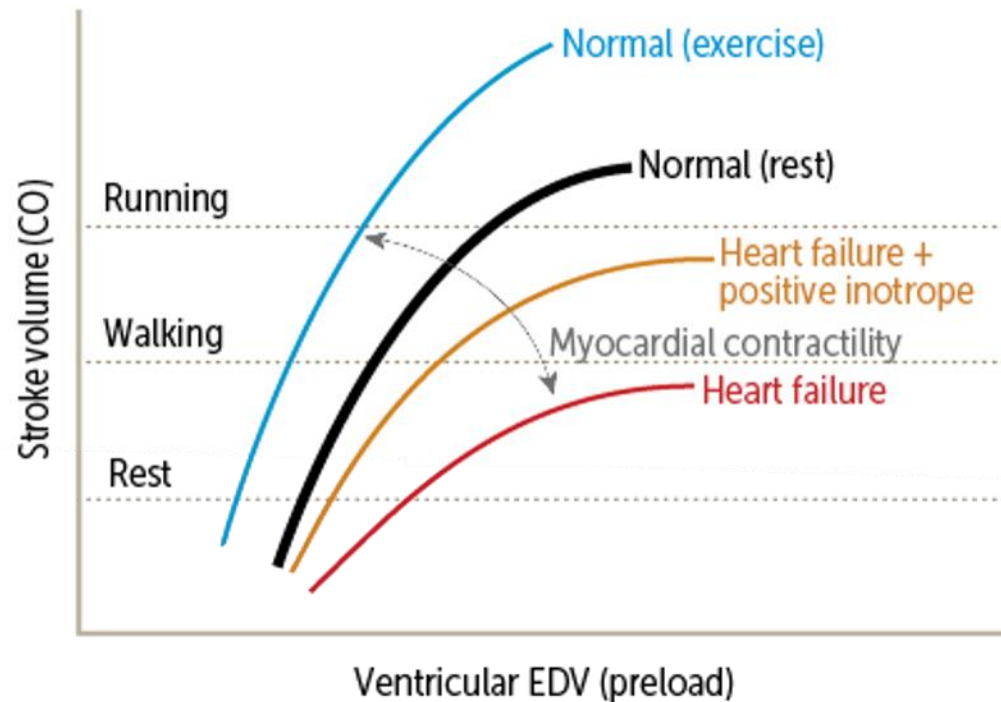


Frank-Starling Law – Effect of Preload (contd)

- **Venous vasoconstriction** contributes to increased **venous return (VR)** and increased **ventricular stretch** → enough force to eject the "extra" returned blood. ESV is not changed.
- This normally **prevents pooling** of blood in pulmonary circulation and systemic circulation (e.g., postural changes).

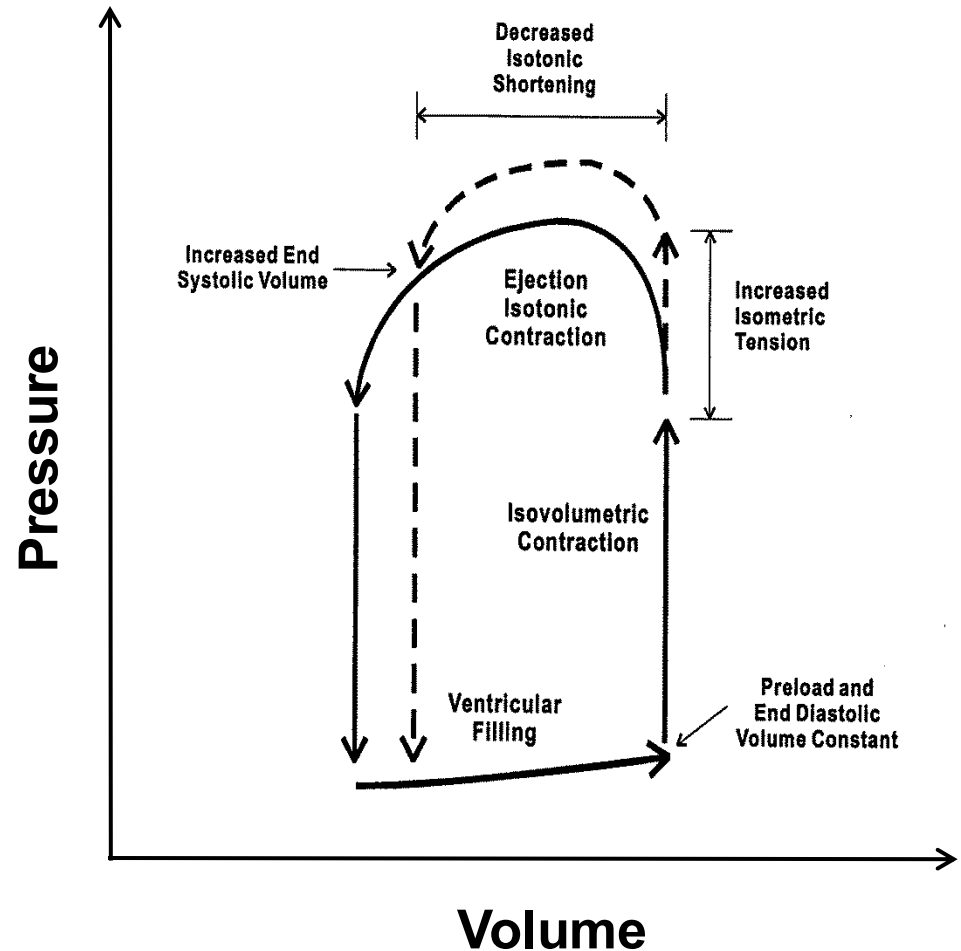


Starling Curves



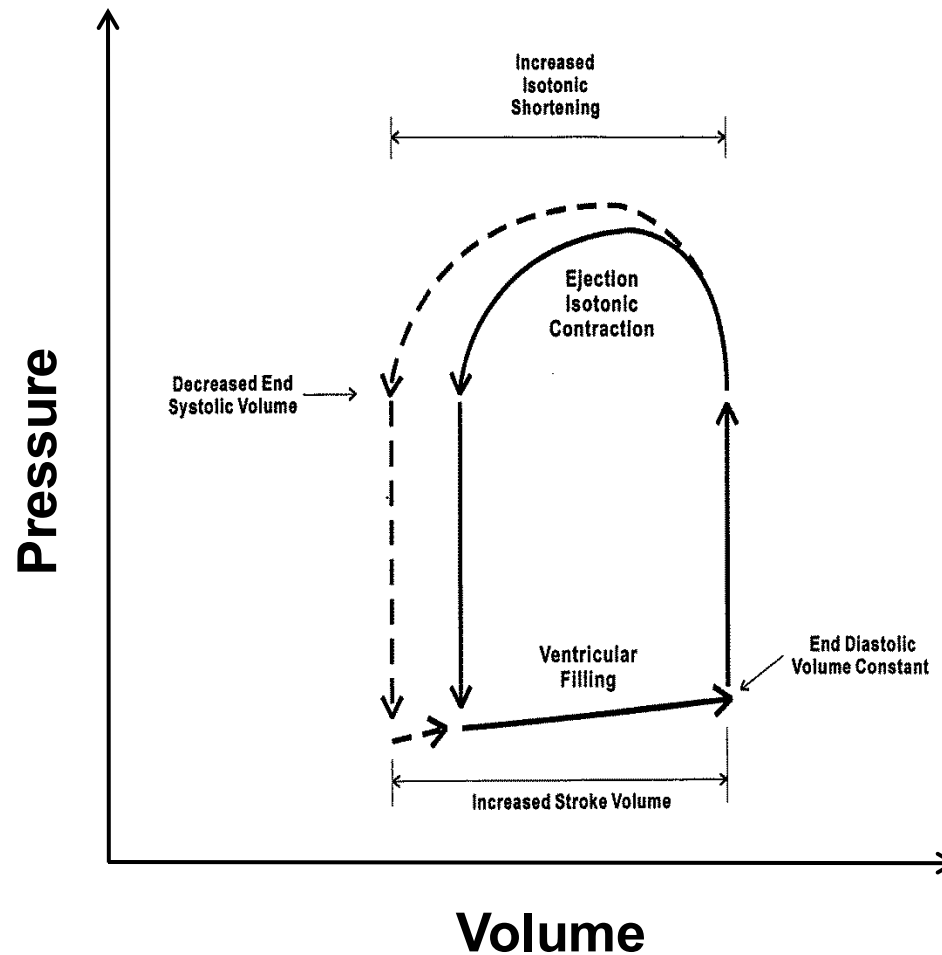
Effect of Afterload

- Afterload is the arterial pressure (AO or PA) that the heart must overcome to circulate blood.
- The greater the afterload, the smaller the SV. The decrease in SV causes an increase in ESV (e.g. more blood remains in the heart) culminating in **heart failure (HF).**
- Hypertension can increase afterload and decrease SV and thus CO.

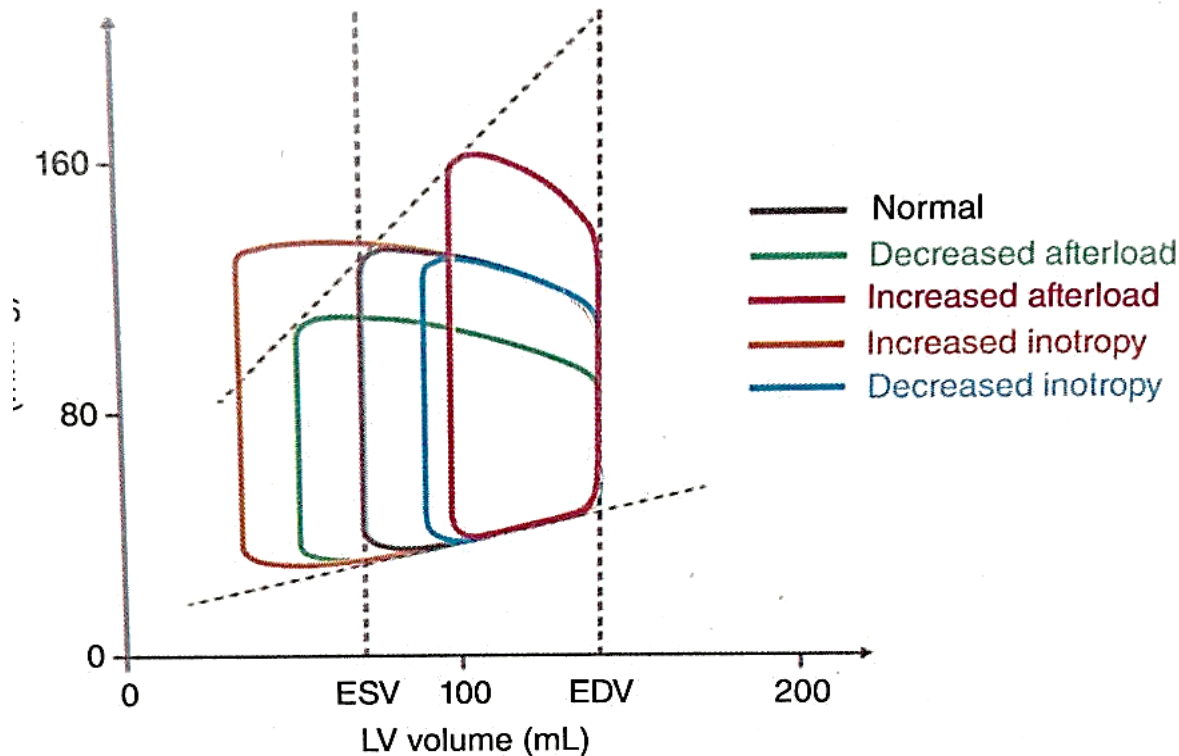


Changes in Contractility – Effect of Norepinephrine

Contractility is an **intrinsic** measure of cardiac performance augmented by factors like Norepinephrine



PV loop in physiological and pathological conditions



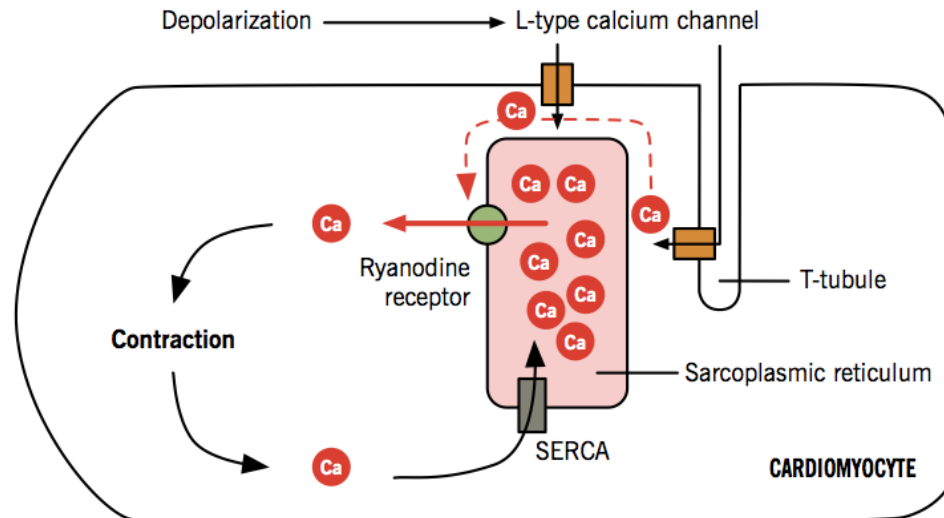
Source: *Comp Physiol* 2016;6:Jan

Changes in Contractility – Calcium

Calcium-induced calcium release

Grigoriy Ikonnikov and Eric Wong

After a depolarization via sodium channels, calcium enters the cardiomyocyte via L-type calcium channels (**orange**). Calcium then activates **ryanodine receptors** (**green**) on the sarcoplasmic reticulum (SR). This special receptor senses intracellular calcium and triggers calcium release from the SR to further increase calcium availability in the cell. As the contraction ends, intracellular calcium returns to the SR via the SERCA calcium channel (**grey**).



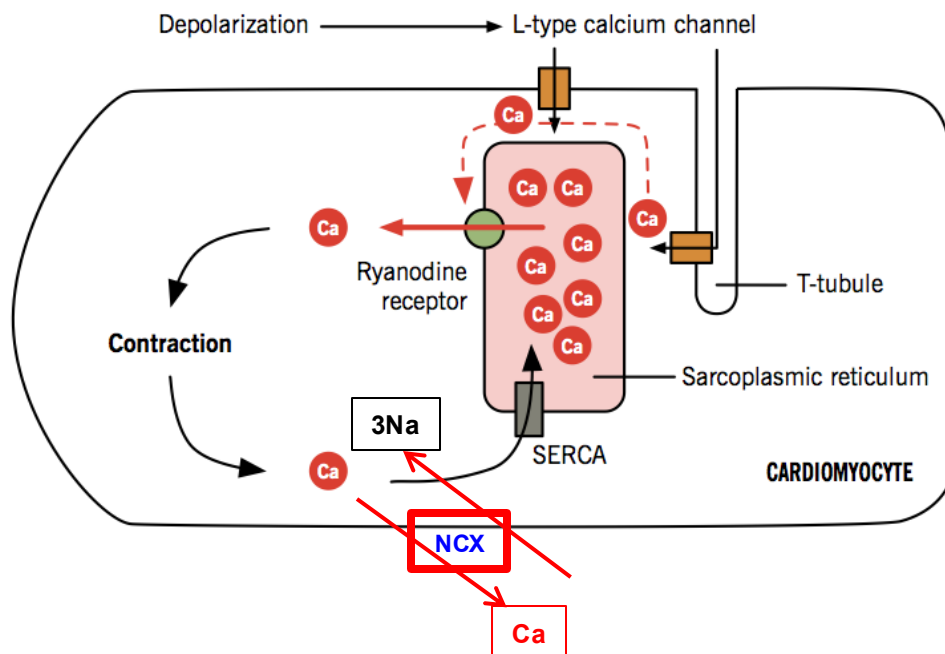
- **Arrival of an action potential** causes an **increase in cytoplasmic Ca⁺⁺** ion concentration which **activates contraction**.
- Sources of Ca⁺⁺ in cardiac muscle include:
 1. **Extracellular fluid**
 2. **Sarcoplasmic reticulum (SR)**

Changes in Contractility – Calcium (Contd)

Calcium-induced calcium release

Grigoriy Ikonnikov and Eric Wong

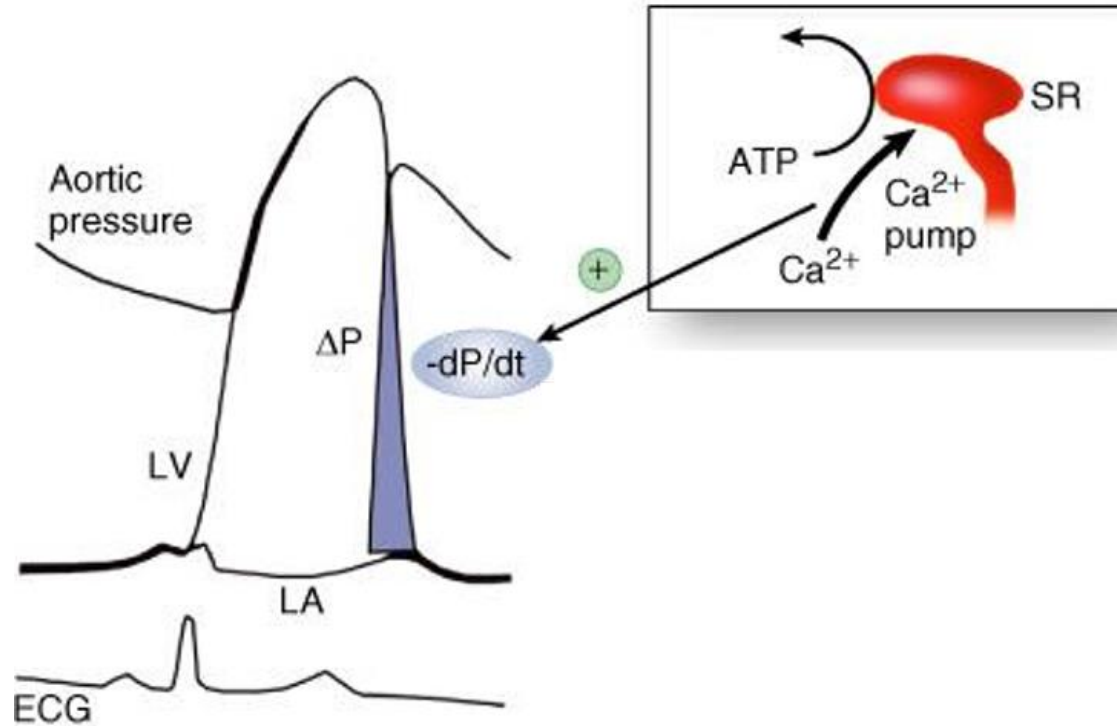
After a depolarization via sodium channels, calcium enters the cardiomyocyte via L-type calcium channels (orange). Calcium then activates **ryanodine receptors** (green) on the sarcoplasmic reticulum (SR). This special receptor senses intracellular calcium and triggers calcium release from the SR to further increase calcium availability in the cell. As the contraction ends, intracellular calcium returns to the SR via the SERCA calcium channel (grey).



Removal of Ca^{2+} ions from the cytoplasm

1. **SERCA** - sarcoplasmic endoplasmic reticulum **Ca^{2+} ATPase** - transports Ca^{2+} ions back into the SR
2. **$\text{Na}^{+}/\text{Ca}^{2+}$ exchanger** uses energy (ATP) released by the movement of one solute (3 Na^{+}) down its electrochemical gradient (e.g., into the cell) to transport a different solute (1 Ca^{2+}) against its gradient or uphill (e.g., out of the cell).

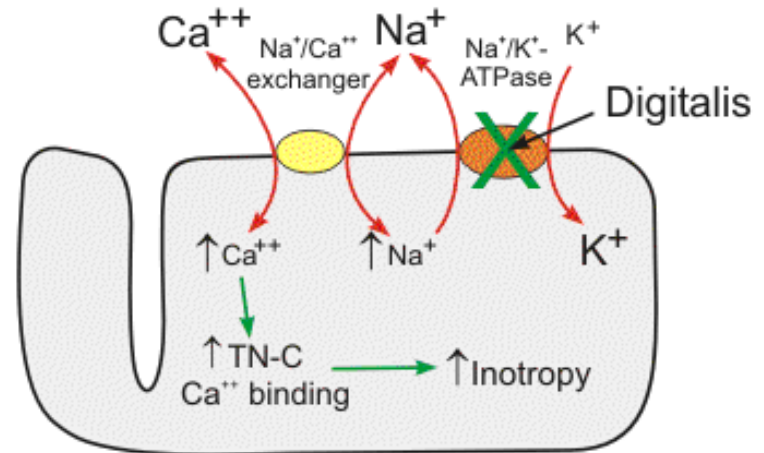
Changes in Contractility – Calcium (Contd)



Modified from Opie LH: Heart Physiology, from Cell to Circulation. Philadelphia, Lippincott Williams & Wilkins, 2004.

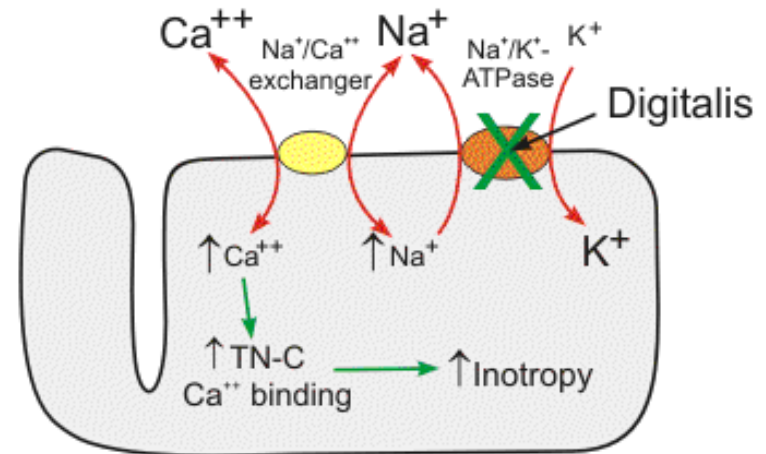
Contractility – Glycosides and Calcium

- In **Heart Failure** (HF), the heart becomes weak due to various causes e.g., Hypertension or myocardial infarction.
- Cardioactive **glycosides** (e.g., **Digoxin**) - inhibitors of the **Na⁺, K⁺ ATPase** or sodium pump.
- **Normal Sodium pump function:**
 - **K⁺ in; Na⁺ out** - ATP-dependent ion transport - **against their electrochemical gradients**.



Contractility – Glycosides and Calcium (Contd...)

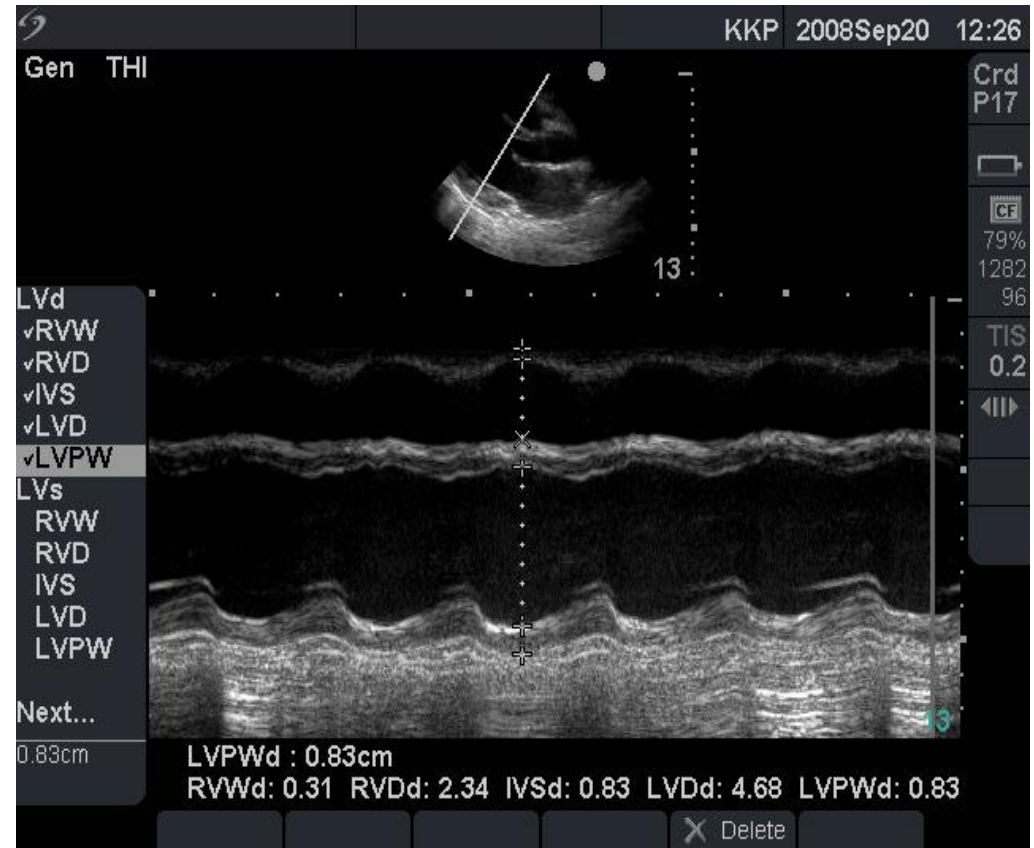
- **Sodium pump with Digoxin - the intracellular Na^+ ion concentration increases.**
- The increase in intracellular Na^+ slows down the $\text{Na}^+/\text{Ca}^{++}$ exchanger (Na^+ in; Ca^{++} out)
- Slowing the $\text{Na}^+/\text{Ca}^{++}$ exchanger increases the cytoplasmic Ca^{++} ion concentration → Force of contraction is increased



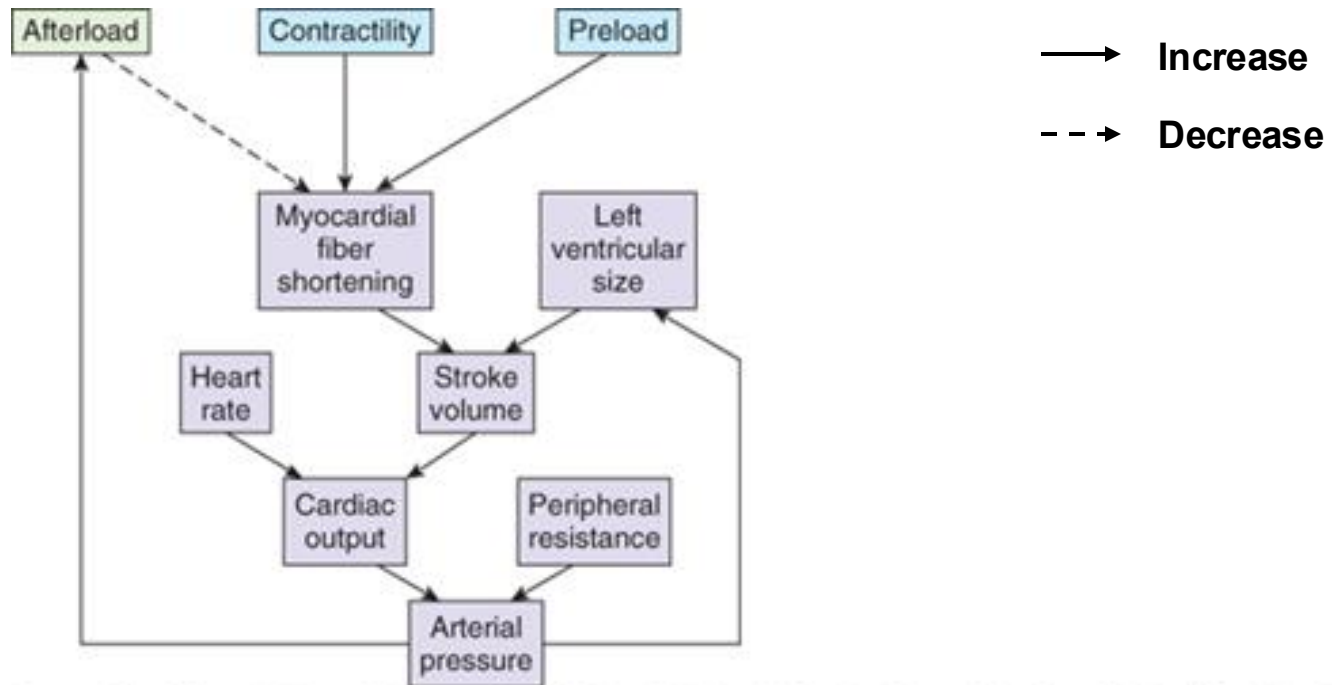
Ejection Fraction (EF; %)

$$= \frac{(EDV - ESV) * 100}{EDV}$$

$$= \frac{SV * 100}{EDV}$$

$$= \frac{(\text{LVIDd} - \text{LVIDs}) * 100}{\text{LVIDd}}$$


Interactions between components regulating CO and arterial BP



Source: Kim E. Barrett, Susan M. Barman, Scott Boltano, Heddwen L. Brooks: Ganong's Review of Medical Physiology, 25th Ed.
www.accessmedicine.com
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Sample question

According to Starling's Law of the heart, cardiac output is directly related to:

- a) the size of the ventricles
- b) the heart rate
- c) amount of blood returning to the heart
- d) end-systolic volume
- e) cardiac reserve

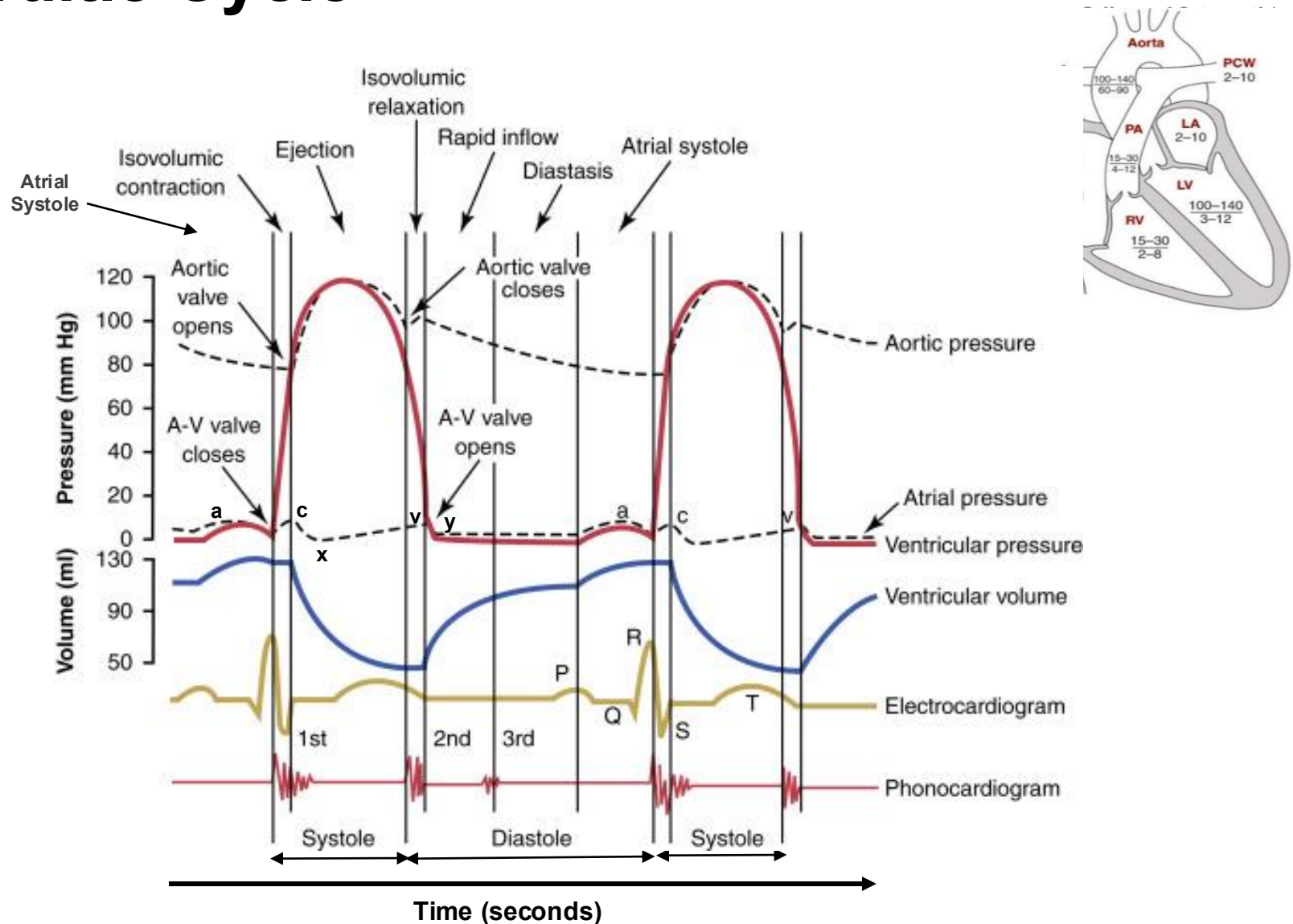
Cardiac Output variables

- Factors that change **stroke volume** include:
 - **P**reload (Frank-Starling Law)
 - Venous vasodilators ↓ preload
 - **A**fterload (~ MAP)
 - Arterial vasodilators ↓ afterload
 - ACE inhibitors and ARBs ↓ preload and afterload
 - Chronic hypertension leads to LV hypertrophy
 - **C**ontractility (intrinsic)
 - beta1-blockade, HFrEF, acidosis, ↓PO₂/↑PCO₂, Non-dihydropyridine Ca⁺⁺ channel blockers ↓ contractility
- **Myocardial O₂ demand** is directly proportional to:
 - **C**ontractility
 - **A**fterload
 - Heart **R**ate
 - **D**iameter of ventricle (wall tension)
- **Ejection fraction**
 - ↓ HFrEF
 - Preserved/Normal in HFpEF

Case study

- A 73-year old man visits ER with complaints of exertional dyspnea, syncope, chest pain. Poorly managed hypercholesterolemia with 11-year history of hypertension.
 - BP: 101/75.
 - Lungs clear. Systolic murmur loudest over the aorta; peaks at mid systole.
 - ECG: LAD, ST normal.
 - CXR: large LV, calcification of aortic valve.
 - Echo: Greatly increased velocity of flow during the systolic portion of the cardiac cycle. Left heart enlarged and LVH.
 - Cardiac enzymes normal.
-
- a) Anti-hypertensive therapy
 - b) Coronary artery bypass grafting
 - c) Percutaneous coronary intervention
 - d) Aortic valve repair/replacement
 - e) Sublingual nitroglycerin

Cardiac Cycle



Source: Guyton and Hall Textbook of Medical Physiology

Summary Review

- **Law of Laplace**
- **Ventricular Pressure-Volume Curve, DCM**
- **Cardiac Output**
- **Regulation of Stroke Volume**
 - **Preload (Frank-Starling Law)**
 - **Afterload**
 - **Contractility and Calcium**
- **Clinical Conditions – MI, valvular pathology**

Interactive Learning of PV Loops with Sample Clinical Questions and Explanations

https://webmedia.unmc.edu/eLearning_open/RFA5/The_Pressure-Volume_Loop/story_html5.html

More sample questions for practicing

1. <https://drive.google.com/file/d/1XBNAGbhbRnIVyGtPFaRnR-ygLB1RUAsZ/view?usp=sharing>
2. <https://exchange.scholarrx.com/brick/contractility-preload-and-afterload>
3. <https://exchange.scholarrx.com/brick/pressure-volume-loops>
4. <https://exchange.scholarrx.com/brick/measuring-cardiac-function>

Detailed Case Studies for Practicing Exam Items

Cardiac Cycle - Parts I and II

(may need NYIT login and/or accessing from the
campus)

- **Cases and Concepts Step 1: Pathophysiology Review (Karen Spizer)**
 - Chapter 1 (Systolic and Diastolic Dysfunction)
 - <https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=49413268&bookid=732>
- **Lippincott® Illustrated Reviews: Physiology**
 - 17: Cardiac Mechanics
 - <https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=258031265&bookid=3310#258031454>
- **Physiology: Cases and Problems**
 - Chapter 2: Cardiovascular Physiology
 - **Search for “cycle” using Ctrl+F or Command+F keys and start reviewing them**
 - https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=49980260&bookid=811#cost4_c002s004cs001

References

- **Guyton and Hall, Textbook of Medical Physiology, 14th and 15th Eds.**
 - Copy & paste should work; direct-click may not work sometimes
 - Online:
 - <https://nyit.idm.oclc.org/login?url=https://www.clinicalkey.com/#!/browse/book/3-s2.0-C20170004883>
 - <https://www-clinicalkey-com.nyit.idm.oclc.org/#!/browse/book/3-s2.0-C20220008424>
 - Chapter 9 - Cardiac Muscle; The Heart as a Pump and Function of the Heart Valves
 - Chapter 20 - Cardiac Output, Venous Return, and Their Regulation
- **ScholarRx Brick(s) and link(s):**
 - Contractility, Preload, and Afterload:
 - <https://exchange.scholarrx.com/brick/contractility-preload-and-afterload>
 - Pressure-Volume Loops:
 - <https://exchange.scholarrx.com/brick/pressure-volume-loops>
 - Measure Cardiac Function:
 - <https://exchange.scholarrx.com/brick/measuring-cardiac-function>



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“The heart of animals is the foundation of their life....., that upon which all growth depends, from which all power proceeds.”

William Harvey (~1628)

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>